

Inferring hatchery effects using spawner-recruit data: comment on Courter et al. (2022)

Matthew R. Falcy 

US Geological Survey, Idaho Cooperative Fish and Wildlife Research Unit, Department of Fish and Wildlife Sciences, University of Idaho, Moscow, ID 83844, USA

Corresponding author: Matthew R. Falcy (email: mfalcy@uidaho.edu)

Courter et al. (2022) address one of the most complicated and disputed topics in fisheries management—the effects of hatchery-origin fish on natural-origin fish. I contend that Courter et al. misinterpret and misstate evidence, adding confusion to a topic that needs clarity.

Using a single-population spawner-recruit analysis, Courter et al. look for an effect of the proportion of hatchery-origin spawners (pHOS) and proportionate natural influence (PNI) on annual deviations in adult recruitment. Finding no effect, Courter et al. state:

“the primary metrics for inferring genetic risks of hatchery programs in the Pacific Northwest (Paquet et al. 2011) were not associated with changes in adult winter steelhead productivity.”

This adds confusion because metrics of genetic risk such as pHOS and PNI are not expected to predict annual deviations in adult recruitment. The metrics pHOS and PNI are used to index genetic risk from domestication selection and introgression that can, over longer periods of time, reduce the productivity of natural-origin spawners. Thus, the Courter et al. modeling and language confuse evolution with demography. A potential demographic outcome from approximately 60 years of domestication selection and introgression is low intrinsic productivity of the naturally spawning population. Indeed, low intrinsic productivity is precisely what Courter et al. find ($\alpha = 2.0$; see Chilcote et al. 2013 for the range of values of steelhead intrinsic productivity), yet their language suggests there is no negative hatchery effect.

The concern over hatcheries is whether *natural-origin spawners* have reduced *productivity*. Courter et al. confuse matters when they state:

“We did not find evidence that hatchery supplementation negatively affected natural-origin steelhead *recruitment* during the 27-year period of record.” (*italics added*)

If hatchery-origin spawners have offspring that survive to adulthood, then such offspring are counted as natural-origin recruits. Thus, we could begin a hatchery program in a basin that does not have steelhead and subsequently find natural-origin recruits. We could also functionally extirpate a historical population and still have natural-origin recruits. The

Courter et al. result quoted above is neither surprising nor informative about the concern over hatcheries. However, the statement misleads readers to believe there is evidence of absence of a problem.

Spawners are a mixture of natural- and hatchery-origin fish. The Courter et al. paper does not cleanly separate recruitment coming from these two groups. Suppose 50 hatchery-origin and 50 natural-origin spawners produce 200 adult recruits. Natural-origin productivity could be $200/50 = 4$ if hatchery-origin spawners produce no adult recruits; natural-origin productivity could be $100/50 = 2$ if hatchery-origin spawners produce equal number of recruits as natural-origin spawners. The Courter et al. analysis does not thus address the productivity of natural-origin spawners per se, even though the language suggests otherwise. If natural-origin spawner productivity has been reduced to the point where it approximates hatchery-origin spawner productivity in the wild (a concern), then we would expect the Courter et al. result of no effect of pHOS on annual deviations in adult recruitment.

The genetic effects of domestication selection and introgression at the epicenter of concern for hatcheries cannot be adequately addressed by the single-population spawner-recruit analysis conducted by Courter et al. To make claims offered by Courter et al., an analysis is needed across populations that correlates productivity and hatchery legacy, while controlling for habitat quality and reduced relative reproductive success of hatchery-origin spawners. This difficult work was not attempted.

I posit that Courter et al. confuse important distinctions between evolution and demography, and between recruitment and productivity of natural-origin spawners. The Courter et al. thesis of no negative effect from hatcheries is not supported by their analysis. Indeed, Courter et al.'s results suggest the opposite thesis: No effect of pHOS on annual deviations in recruitment and low population productivity are consistent with a negative effect of hatchery fish on productivity of natural-origin spawners. How did we get here? First, there is a linguistic matter that needs to be cast in technical terms. Placing an annual covariate on intrinsic productivity

(α) causes the entire spawner-recruitment function to move up or down with annual deviations in the covariate (α is in the equation for the stable equilibrium abundance). I herein refer to the effect of the covariates (namely pHOS) as operating on “annual deviations in adult recruitment” because the Courter et al. modeling technique functions similarly to the more traditional method of using the covariate to explain residual variation in the response variable (adult recruitment).

There are many other issues in Courter et al. not discussed in this commentary (priors not described, misinterpretations of others’ published work, type II inferential error). While some specialized scientists can judge the severity of all the issues in Courter et al. for themselves, members of the broader public usually cannot. Specious language that sews confusion about evidence for effects (or lack thereof) of hatchery-origin fish allows spin and motivated reasoning to affect public policy. The limits of scientific inference from spawner-recruit data require broader understanding among scientists so that public policy about hatcheries can reflect a proper synthesis of risk and reward.

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Author information

Author ORCIDs

Matthew R. Falcy <https://orcid.org/0000-0002-3332-2239>

Author contribution

Conceptualization: MRF

Writing – original draft: MRF

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